

Emissions From the Exhaust of FCIPT's Plasma Pyrolysis System

Pollutant	CPCB Standard*	Plasma System
CO	100 mg/Nm ³	40-85 mg/Nm ³
NOx	400 mg/Nm ³	7-25 mg/Nm ³
PM	150 mg/Nm ³	31-105 mg/Nm ³
Dioxin & Furan	0.1 ng/Nm ³ TEQ	0.1 ng/Nm ³ TEQ

*Central Pollution Control Board's emission standards
Source: www.plasraindia.com

potential is likely to increase further with economic development.

Indian domestic wastes contain 60-80 per cent moisture and 70-80 per cent biodegradables and 7-10 per cent combustibles by wet weight. In comparison, in the West, with relatively dry wastes, the reverse is true: They contain 16-24 per cent biodegradables and the rest are mostly combustibles. Hence, burn technologies are unviable in India.

Plasma gasification plants. Presently, there are two 68 tonnes-per-day hazardous waste disposal plants utilising the WPC's PGVR technology. As mentioned earlier, these are being constructed at Pune and Nagpur by SMS Infrastructures Limited to produce 5 MW of electricity.

Plasma pyrolysis system designed by FCIPT. The Gujarat-based Facilitation Centre for Industrial Plasma Technologies (FCIPT), which is a part of the Institute of Plasma Research, in collaboration with TIFAC of the Department of Science and Technology, has designed a plasma pyrolysis system.

The torch (consisting of two electrode devices that convert electrical energy into heat energy) has also been developed by FCIPT. In the primary chamber, waste is fed and treated with plasma emitted by the torch. Thereafter, gases and slag are formed in the chamber. The gases include carbon monoxide, nitrogen and hydrocarbons. These are decontaminated (scrubbed) in the secondary chamber (with the help of a chemical/water shower). The temperature of the gases is also reduced from 600°C to less than 80°C

during the quenching process. The entire process takes place in the absence of oxygen.

The system can treat 15-20 kg of hospital waste per hour. It can be set up at a cost of Rs 1-1.2 million. The plasma pyrolysis method can be used efficiently for municipal solid waste and

plastic waste also. The gas product has a heating value of approximately one-third the heating value of natural gas. Therefore it can be used as an efficient fuel source for industrial processes, including electricity generation and production of methanol and ethanol. The slag can be used in the construction industry or for road paving. Plasma discharge treatment has no byproducts which must be disposed of as waste. Therefore it can be viewed as a totally closed treatment system. The plasma pyrolysis system is shown in Fig. 10.

Four systems have been supported by the Department of Science and Technology for field installations for plastic/municipal waste disposal, focusing on the demonstration in ecologically sensitive regions.

Meanwhile, Bhabha Atomic Research Centre (BARC) has provided technology for conversion of kitchen wastes, stale food, split milk, leftovers from hotels and vegetable refuse into organic manure, offering excellent topsoil material to the farmers. A few such plants have already been installed in Maharashtra, Orissa and Gujarat.

Similarly, Mumbai-based Asian Electronics Ltd and Singapore's Enviro-Hub Holdings have teamed up to build four power plants of 8 MW each using a commercially viable technology for producing electricity from plastic waste.

Here are the excerpts of an e-mail posted by Hareesh Iddya of Indian Institute of Science (IISc), Bangalore, on May 4, 2007 on 'Down to Earth' website, which show some

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